

Counting Mangroves from Space: Weakly Supervised Canopy Mapping with Sentinel-2

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ABSTRACT

Mangrove maps are usually drawn by hand every few years, while coastlines change every season. We train a segmentation model on Sentinel-2 imagery using weak labels from outdated national maps and a small set of 320 carefully checked tiles. A noise-aware loss and seasonal compositing push the F1 score from 0.71 to 0.86 on an independent field survey. The resulting maps show 3.4 percent net canopy loss across the study coast over four years, concentrated around new aquaculture ponds.

Keywords: mangroves, Sentinel-2, weak supervision, semantic segmentation, coastal monitoring

1. Introduction

Mangroves protect coasts, store carbon and feed fisheries, yet the maps we use to protect them are often older than the ponds that replaced them. Satellites see the coast every five days; the labels are the bottleneck.

2. Method

We composite cloud-free median images per season and compute a few classic indices next to the raw bands. Old maps provide cheap but noisy labels. A small clean set lets us estimate how noisy they are, and a forward-corrected loss downweights pixels the old maps probably got wrong [2].

The network is a compact U-Net that fits on a single consumer GPU, so regional agencies can run it.

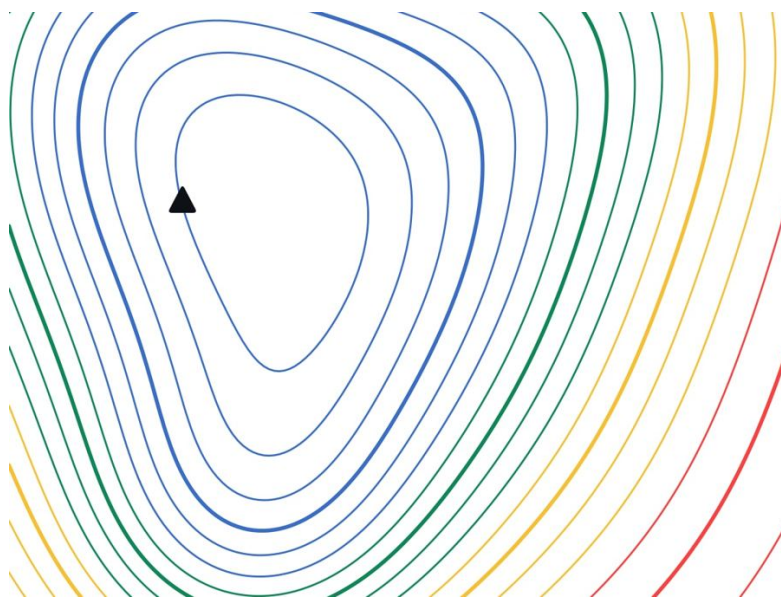


Figure 1. Canopy probability contours for one estuary. Warm lines mark high confidence mangrove canopy.

3. Results

Noise-aware training gives the largest single improvement, and seasonal composites help mostly in cloudy estuaries (Table 1). Change maps agree with field notes at 17 of 20 visited sites.

Training setup	Precision	Recall	F1
Old maps only	0.74	0.68	0.71
+ clean tiles	0.81	0.77	0.79
+ noise-aware loss (ours)	0.87	0.85	0.86

Table 1. Accuracy against the independent field survey.

4. Discussion

The model still confuses young mangroves with other coastal shrubs. Adding radar backscatter is the obvious next step because it sees structure, not only color.

5. Conclusion

Weak labels plus a small clean set are enough for useful, repeatable mangrove maps.

References

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